



OLEDing the way for TV panels...



hen it comes to flat-screen TVs our options so far were limited to plasma, CCFL backlit LCD and LED backlit LCD panels.

This year should see an addition to that in the form of OLED panels. But the only catch is that OLED TVs are going to be prohibitively expensive when they launch - LG has already started with pre-orders for its curved screen 55-inch OLED TV which will retail at USD 13,500. And no 'curved' wasn't a typo. We'll get to that in a bit. Organic Light Emitting Diodes or OLED has already made its entry on the smart-phone screens in the form of AMOLED displays - think Samsung Galaxy S2/S4 or Nokia Lumia 820 or even Microsoft Zune. OLEDs are capable of generating their own light when activated with an electrical current and do not require any sort of fluorescent or LED backlight. We have already seen the difference in terms of thickness between an LED backlit LCD TV and a CCFL backlit LCD TV - both kinds require backlighting. So when you replace these with an OLED screen which

does not require any kind of backlight you get TVs which are slimmer than the slimmest LED backlit LCD TV. LG gave a glimpse of that in CES 2012 where it showcased its 55-inch OLED TV which was just 4mm thick! The next immediate advantage is the bump in contrast levels - thanks to the lack of backlighting, OLED panels produce much deeper blacks and are they are also capable of bright whites. Another obvious advantage is reduced power consumption. Coming back to the curved aspect - an OLED is made out of an organic or polymer based substrate which



LG's curved OLED TV which is retailing for \$13,500

can be flexed or bent without affecting the flow of current or changing its light-emitting properties. At CES 2013 Samsung announced its flexible OLED prototype products calling these displays YOUM displays. More recently at the Society for Information Display (SID) week, LG showcased its 5-inch unbreakable and flexible plastic OLED panel which could be used in future mobile displays. Another prototype was showcased by Sony in 2010 which shows a flexible OLED display being rolled around a tube the size of a pencil while it was playing back a low res video. The possibilities are endless.

Sure the cost will be restrictive, but then HDTVs were also prohibitively expensive when they first came onto the scene. LG, Samsung, Sharp have already shown their OLED prowess at electronics fairs over the years. Sony and Panasonic have decided to team up for OLED TV production to give competition to the Korean giants LG and Samsung. Competitive pricing will get OLED TV prices to sensible levels. When that will happen? Well, as of now that is anyone's guess.

Pixel density, where does it end?



ver since Apple introduced the Retina display with the iPhone 4, pixel density as a specification has acquired a pride of place in marketing

jargons. What started on phones only, is now also used to differentiate tablets as well as laptops. But is it just a gimmick or should it be a prime consideration before you get your mobile phone or tablet or laptops? If you don't already know how pixel density is calculated, or if you're too lazy to do the math head to pixeldensitycalculator.com - just fill up the details and get the PPI for any screen. Increasing the number of pixels on your device will improve the quality no doubt,

but beyond a point the human eye cannot really resolve individual pixels. For instance taking Apple products the pixel density for iPhone 4/4S is 326ppi, third generation iPad is 264ppi and that of the 13-inch MacBook Pro is 227ppi. The numbers are different for these devices as they are used at different distances. The phone is used at a closer distance from the eye, hence a higher pixel density. Tablets are used at a comparatively farther distance hence a lower pixel density. If you thought Apple's numbers were high - the 5-inch HTC One has a whopping 469ppi and Samsung has announced a 13.3-inch display with a 267ppi which equates to a resolution of 3200x1800 pixels! Better

display is a given and you will no longer see pixellation around the edges of sharp lines or text or images. Things will appear much more smoother. But where does the buck stop? Higher pixel density means that the processors which power these devices have to pump in that many pixels which means more work, which in turn means more battery consumption. But that is just one part of the problem. If you have a high pixel density device, but the programs or applications you are running are not built keeping that in mind the investment can be pointless. It's kind of like running stretched iPhone apps on the iPad. This creates a challenge for the developers as they need to ensure that